

Cahit Erden Anıl - 572

Homework,

- Outline:
- ① Various examples on S.V.K Thm with some preliminary notion all selected from HW questions.
 - ② Proof of Brouwer fixed point thm using L.E.S.

I will provide some applications of
 ① S.V.K. theorem which gives as
 exercises on notes.

at first glance, applying S.V.K. on CW
 complexes do not seem to be saying
 much as one can "hardly" guess
 the fundamental groups of such
 surfaces. I may give some examples,

$$\pi_1(\text{torus}) = \pi_1\left(\text{box with } \begin{matrix} \xrightarrow{a} \\ \downarrow b \end{matrix}\right) = \{a, b : \underbrace{ab a^{-1} b^{-1}}_{=1}\}$$

$$\pi_1(\text{pair of pants}) = \pi_1(\text{6-gon}) = \{a, b, c, d : [a, b][c, d] = 1\}$$

*: We saw this explicitly in class. For the

$g=2$ case choose $A = e_\varepsilon^2$ (open)

$B = \Sigma_{1,2} - e_{<\varepsilon}^2$ yielding intersection is

some annulus between them.

Then the amalgamated product gives

1 = boundary of our CW-complex with "some" orientation.

AND, 1 comes from triviality of the fundamental group of convex, $\mathbb{R}^2$.

So far, I deliberately talk loosely with the assumption reader is familiar with S.V.K and amalgamated product.

My aim, is to show it for non-tricky in my understanding, not easily be written as CW-complexes!

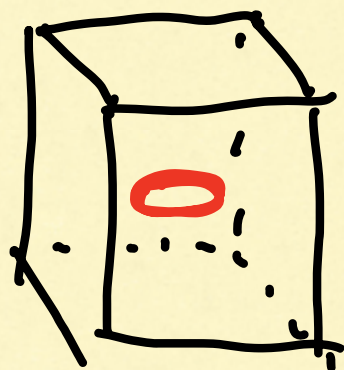
(Also, observe that it is also true for non-orientable CW-complexes)
the way i use the, CW-complexes do not change, even for the case $\mathbb{R}P^2$, Klein bottle.

Just see $\mathbb{RP}^2$ to $v. \circlearrowleft v.$,
rather than $v \begin{array}{c} \rightarrow w \\ \uparrow \downarrow \\ w \rightarrow v \end{array}$.

For the Klein bottle usual CW-complex
suffice.

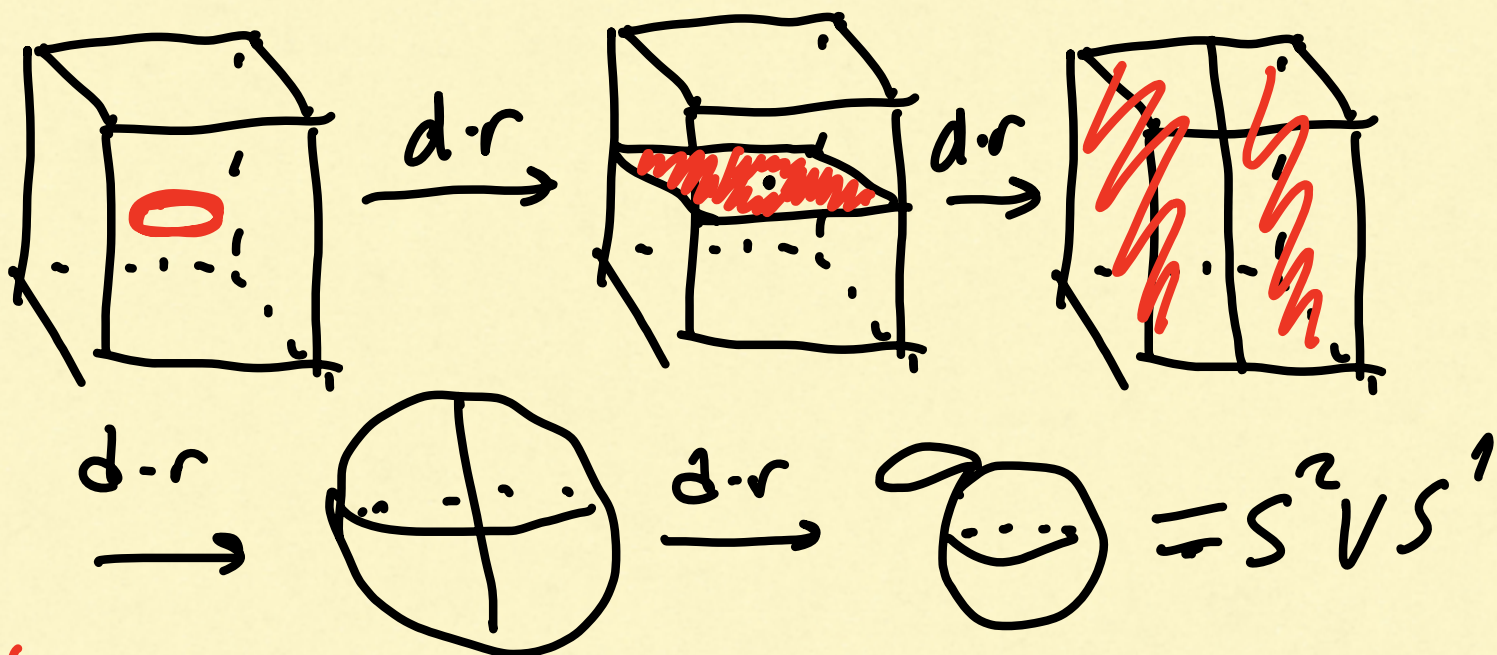
To have some perception about
which spaces i am talking about
let us do some examples.

(i) $\mathbb{R}^3 - \{0\}$

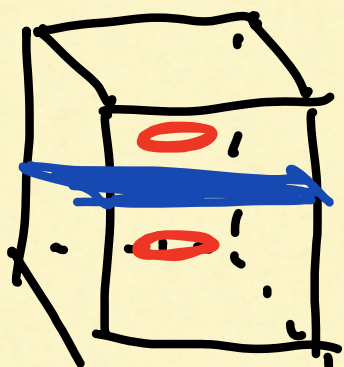


in order to some idea
what it deformation retracts,
recall that plane $\rightarrow \text{d.r.}$
to boundary of the plane.
plane $\rightarrow \text{d.r.}$ to
boundary of the plane and
with some loop based at vertex.

Under this intuition, one could see that



Φ : What about $\mathbb{R}^3 - \bigcirc \cup \bigcirc$?



See as a union of two spaces of the previous example. Intersection is some D^2 with trivial fundamental group.

$$\text{then } \pi_1(X) = \pi_1(A) \underset{\pi_1(A \cap B)}{*} \pi_1(B) \rightarrow \text{gives no relation!}$$

$$= \pi_1(S^1) * \pi_1(S^1)$$

$$= \mathbb{Z} * \mathbb{Z}$$

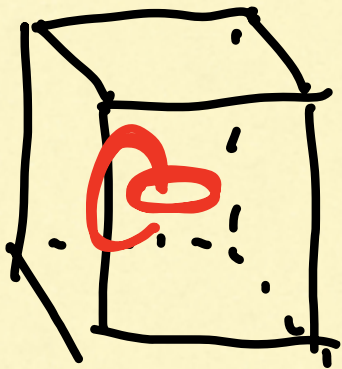
as any $S^{n \geq 2}$ has trivial fundamental group.

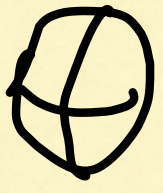
To see that just project $S^{n-2} \hookrightarrow \mathbb{R}^{n-2}$ to $\mathbb{R}^{n-2}$ with stereographic projection.

$$\pi_1(S^n) = \underbrace{\pi_1(\mathbb{R}^n)}_{\cong \pi_1(\mathbb{R}^n)} * \underbrace{\pi_1(\mathbb{R}^n)}_{\cong \pi_1(\mathbb{R}^n)} = 0.$$

our amalgamated presentation do not matter as both is trivial!

Φ : $\mathbb{R}^3 \rightarrow \mathbb{R}^3$.



Boundary of cube $\xrightarrow{\text{dir}}$ 
Now, we have solid S^2
 with link erased.

Knot on right erases the interior of solid S^2 .
 with not erasing the outer any part
 of left erased knot, i.e. left
 knot erased knot becomes a TORUS!
 wedged to our S^2 !

$$\Rightarrow \text{[A sphere with a grid of lines]} = S^2 V T^1$$

Now as one can guess wedge do not give us any information since it gives in a "superficial way", i.e. at intersection point there is always some neighborhood with trivial fundamental group. S.V.K yields

$$\pi_1(\mathbb{R}^3 - \hookrightarrow) = \pi_1(S^2 V T^1) = \{a, b: [a, b] = 1\}$$

②

Q:

A great example on how L.E.S gives us opportunity to find some homology groups and its relation with some of the known thm "Brouwer fixed pt thm".

A: If you recall the proof of B.f.p.t, we construct some continuous function that maps $D^n \rightarrow \partial D^n$. Construction is somehow related on the book. For the homology case it is indeed similar but with the spice "functor" property s.t $(fg)_* = f_* g_*$, $\pi_* = \pi$.

Let us show there is no continuous retraction $D^n \rightarrow \partial D^n$.

if so say this $r: D^n \rightarrow \partial D^n$.

For the usual inclusion $j = \text{id}_{\partial D^n}$,

$$\text{and } (r)_* = [\text{id}_{\partial D^n}]_* = \text{id}_{H_n(\partial D^n)}!$$

before we proceed observe that

$(D^n, \partial D^n)$ is indeed a "good pair", as

∂D^n some sub-complex.

To be able to write some meaningful
L.E.S we should recall that,

$$H_k^{\sim}(S^n) \xrightarrow{\partial} H_{k-1}^{\sim}(S^{n-1}) \text{ isomorphism}$$

from the pair (D^n, S^{n-1}) and exactness
property. $\Rightarrow H_k^{\sim}(S^n) = 0, H_k^{\sim}(S^k) = \mathbb{Z}$.

Now, we have isomorphism, for any n !

$$H_k(\partial D^n) \xrightarrow{j_*} H_k(D^n) \xrightarrow{r_*} H_k(\partial D^n)$$

with corners are 0 or $\mathbb{Z}$ depending the choice of k .

$$\mathbb{Z} \xrightarrow{i_*} \tilde{H}_k(D^n) \xrightarrow{r_*} \mathbb{Z}.$$

or

$$0 \xrightarrow{i_*} \tilde{H}_k(D^n) \xrightarrow{r_*} 0$$

and still exact sequence, allegedly!

as $\tilde{H}_k(D^n) = 0$ by contractibility,

contradiction ■